

Magnetic Circuit

Solved Examples:

Example (1):

A magnetic flux density of 1.2 wb/m^2 having an iron path 1 m long. Calculate the mmf required, assuming a relative permeability of iron is 1500 .

$$B = 1.2 \text{ wb/m}^2 \quad l = 1 \text{ m}$$

$$\text{mmf} = ? \quad \mu_r = 1500$$

$$B = \mu_r \mu_0 H$$

$$H = \frac{NI}{l} = \frac{\text{mmf}}{l}$$

$$B = \frac{\mu_r \mu_0 \text{mmf}}{l}$$

$$\text{mmf} = \frac{lB}{\mu_r \mu_0}$$

$$= \frac{1 \times 1.2}{1500 \times 4\pi \times 10^{-7}}$$

$$= 636.619 \text{ AT}$$

Example (2):

An iron of 50 cm mean diameter has 1000 turns of insulated copper wire wound uniformly on it.

If it produced a flux of $2.5 \times 10^{-3} \text{ wb}$ when a current of 4 Ampere flows in it. Given that the permeability 782 .

Calculate (i) the cross-sectional area of iron ring (ii) magnetizing current (iii) reluctance.

$$d = 50 \text{ cm} = 0.50 \text{ m} \quad N = 1000 \text{ turns}$$

$$\phi = 2.5 \times 10^{-3} \text{ wb} / I = 4 \text{ Ampere}$$

$$\mu_r = 782$$

(i) A (ii) H (iii) S

$$(i) \text{ Area} = \pi \frac{d^2}{4}$$

$$A = \frac{\pi (0.50)^2}{4} = 0.196 \text{ m}^2$$

$$(ii) H = \frac{NI}{l} =$$

$$l = \text{Circumference} = 2\pi r = \pi d$$

$$= \pi (0.50)$$

$$l = 1.570 \text{ m}$$

$$H = \frac{1000 \times 4}{1.570} = 2546.47 \text{ A}$$

$$(iii) S = \frac{l}{\mu_r \mu_0 A}$$

$$= 1.570$$

$$782 \times 4\pi \times 10^{-7} \times 0.196$$

$$S = 8151.301 \text{ Wb/AT}$$

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Example (3):

A circular magnetic circuit has circumference 15 cm. If the current of 10 mA flowing through the circuit of 700 turns if the permeability of the iron circuit is 999. Calculate:

(i) the magnetizing circuit.

(ii) the total flux

(iii) the reluctance.

$$2\pi r = \pi d = 15 \text{ cm} = 0.15 \text{ m}$$

$$I = 10 \text{ mA} = 10 \times 10^{-3} \text{ Ampere}$$

$$N = 700 \text{ turns} \quad \mu_r = 999$$

(i) Magnetizing circuit

$$H = \frac{IN}{l} = \frac{10 \times 10^{-3} \times 700}{0.15}$$

$$H = 46.67 \text{ AT/m}$$

(ii) $\phi = ?$

$$B = \mu_r \mu_0 H$$

$$= 4\pi \times 10^{-7} \times 999 \times 46.67$$

$$B = 0.0585 \text{ wb/m}^2$$

$$B = \frac{\phi}{A} \text{ wb/m}^2$$

$$\phi = B \cdot A$$

$$\pi d = 0.15 = \frac{\pi}{4}$$

$$d = \frac{0.15}{\pi}$$

$$A = \pi \frac{d^2}{4} = \frac{\pi}{4} \frac{(0.15)^2}{\pi^2} = \frac{(0.15)^2}{4\pi}$$

$$\phi = 0.0585 \times \frac{(0.15)^2}{4\pi}$$

$$\phi = 1.047 \times 10^{-4} \text{ wb} ?$$

Another solution:

$$\text{mmf} = \frac{\phi}{S}$$

$$\phi = \text{mmf} \cdot S = IN \cdot \frac{l}{\mu_r \mu_0}$$

$$\phi = \frac{10 \times 10^{-3} \times 700 \times 0.15 \times 4\pi}{999 \times 4\pi \times 10^{-7} \times (0.15)^2}$$

$$999 \times 4\pi \times 10^{-7} \times (0.15)^2$$

$$\phi = 4.67133 \times 10^{-4} ?$$

$$(iii) S = \frac{l}{\mu_r \mu_0} = \frac{0.15 \times 4\pi}{4\pi \times 10^{-7} \times 999 \times (0.15)^2}$$

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Example (4):

A circular magnetic circuit has diameter of 15 cm if the current flowing through the circuit is 5 mA, where the permeability of iron circuit is 1200

Calculate

- ① mmf
- ② flux
- ③ Magnetizing strength force
- ④ reluctance

$$d = 15 \text{ cm} = 0.15 \text{ m}$$
$$I = 5 \text{ mA} = 5 \times 10^{-3} \text{ Ampere}$$
$$\mu_r = 1200$$

$$\text{mmf} = NI = \frac{\phi}{S}$$

$$B = \mu_r \mu_0 H$$

$$H = \frac{B}{\mu_r \mu_0}$$

$$\text{Area} = \pi \frac{d^2}{4} = \frac{\pi}{4} (0.15)^2 = 0.01767 \text{ m}^2$$

$$l = 2\pi r = \pi d = \pi \times 0.15 = 0.471 \text{ m}$$

$$S = \frac{l}{\mu_r \mu_0} = \frac{0.471}{4\pi \times 10^{-7} \times 1200 \times 0.01767}$$

$$S = 17676.376 \text{ Wb/AT}$$

$$H = \frac{1}{1200 \times 4\pi \times 10^{-7}} B$$

$$H = 663.145 \text{ B AT/m}$$

$$\text{mmf} = H l \quad H = \frac{NI}{l}$$

$$\text{mmf} = 663.145 \times B \times 0.471$$

$$\text{mmf} = 312.341 \text{ B AT}$$

$$\phi = \frac{B}{A} = \frac{1}{0.01767} B$$

$$\phi = 56.59 \text{ B wb}$$

$$\Phi = S \cdot m \cdot f = S \cdot NI$$

$$\Phi = 7177419.355 \times 500 \times 1$$

$$B = \frac{\Phi}{A}$$

$$\frac{wb}{m^2}$$

$$\Phi = 358879678$$

Example (5):

Calculate the AT/m required to produce a flux density of 0.001 wb/m^2 at the center of a long straight solenoid. (given $\mu_r = 900$).

$$AT/m = ?$$

$$B = 0.001 \text{ wb/m}^2$$

$$H = \frac{NI}{L}$$

$$B = \mu_r \mu_0 H$$

$$H = \frac{B}{\mu_r \mu_0} = \frac{0.001}{900 \times 4\pi \times 10^{-7}}$$

$$H = 0.884 \text{ AT/m}$$

Example (6):

A non-magnetic ring has mean diameter of 44.5 cm and a cross-sectional area.

It is uniformly wound with 500 turns.

Calculate the field strength / total flux produced

in the ring by a current of 1 A and the reluctance.

$$d = 44.5 \text{ cm} = 44.5 \times 10^{-2} = 0.445 \text{ m}$$

$$N = 500 \text{ turns}$$

$$B = \frac{\Phi}{A} \Rightarrow \Phi = BA \quad I = 1 \text{ A}$$

$$\text{mmf} = \frac{\Phi}{S} \Rightarrow \Phi = S \text{ mmf}$$

$$B = \mu_r \mu_0 H \quad S = \frac{l}{\mu_r \mu_0 A} \quad H = \frac{NI}{L}$$

$$B = \frac{\Phi}{A}$$

$$H = \frac{NI}{L} = \frac{500 \times 1}{\pi d} = \frac{500 \times 1}{\pi \times 0.445}$$

$$H = 357.651 \text{ AT/m}$$

$$A = \pi \frac{d^2}{4} = \frac{\pi}{4} (0.445)^2$$

$$A = 0.155 \text{ m}^2$$

$$S = \frac{l}{\mu_0 A}$$

$$S = \frac{\pi d}{\mu_0 A} = \frac{\pi \times 0.445}{4\pi \times 10^{-7} \times 0.155}$$

$$S = 7177419.355 \text{ AT}$$

$$\Phi = S \cdot \text{mmf} = S \cdot NI$$

$$\Phi = 7177419.355 \times 500 \times 1$$

$$\Phi = 358879678$$

$$B = \frac{\Phi}{A}$$

$$\frac{\text{wb}}{\text{m}^2}$$

Exercise: -

Example (7)

An iron ring 1m mean circumference is made from iron rod of 200 turns.

what current would be necessary

to produce a flux 1m wb.

and magnetizing current if

the relative permeability is 520.